

Rongqian Will Chen

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Education

George Washington University (GWU)

PhD in Electrical Engineering

University of Pennsylvania (UPenn)

MSE in Electrical Engineering

Southwest Jiaotong University (SWJTU)

BEng in Automation

Washington, D.C., US

Sept. 2024 – Now

Philadelphia, US

Sept. 2021 – May 2023

Chengdu, China

Sept. 2017 – Jun. 2021

Skills

Mathematical Tools Mathematica, Matlab

Design and Simulation SolidWorks, Ansys, Unity, PSIM

Hardware Development Altium Designer, Keil, Proteus, CCStudio

Embedded Systems Linux, Raspberry Pi, Arduino, STM32, FPGA, DSP

Programming Languages Python, C, C++ , C#, Verilog, HTML

Publications

- **Rongqian Chen**, Jun Kwon, Kefan Wu, Wei-Hsi Chen. Tunable Stiffness for Energy-Efficient Vertical Hopping in a Monopedal Robot Across Varying Ground Profiles. *ICRA 2025*.
- **Rongqian Chen**, Jun Kwon, Wei-Hsi Chen, Cynthia Sung. Design and Characterization of a Pneumatic Tunable-Stiffness Bellows Actuator. *RoboSoft 2024*.
- Shivangi Misera, Mason Mitchell, **Rongqian Chen**, Cynthia Sung. Design and Control of a Tunable-Stiffness Coiled-Spring Actuator. *ICRA 2023*.
- **Rongqian Chen**, Yingquan Zou, Anyong Gao, Leshi Chen. A Cluster-Based Weighted Feature Similarity Moving Target Tracking Algorithm for Automotive FMCW Radar. *VTC 2022-Spring*.
- Ping Yang, Xi Chen, **Rongqian Chen**, et al. Stability improvement of Pulse Power Supply with dual-inductance Active Storage Unit Using Hysteresis Current Control. *IEEE JETCAS*, 2021.

Experience

Pneumatic Legged Hopping Robot

Sept. 2022 – July 2024

- Built a legged hopper and a tunable-stiffness pneumatic bellows (no external pressure), achieving a 143% stiffness range via analytic, data-driven, and FEA modeling.
- Architected the mech/elec stack and comms; integrated Raspberry Pi, Pico, Arduino, ESP32, BLDC drive, perception, and data logging.
- Demonstrated a 29.3% reduction in energy loss via stiffness tuning; designed a five-bar ground emulator with stiffness/damping control for validation.

Tunable-Stiffness Coiled-Spring Actuator

Oct. 2021 – Sept. 2022

- Modeled elastic-ring deformation and multilayer coil stiffness; simulated force–displacement curves.
- Designed 3D-printed mechanism, transmission, and electronics; implemented motor control; deployed on a flexible manipulator with motion/path tracking.

Moving Target Tracking Algorithm of mmWave Radar

Nov. 2020 – July 2021

- Built and calibrated a 77 GHz FMCW radar road-test platform.
- Developed signal processing and a robust tracker from sparse radar features to improve resolution and detection robustness.

Pulse Power Converter with Dual-Inductance Active Storage Unit

Sept. 2019 – Aug. 2020

- Designed op-amp control and refined hysteresis current control, reducing current spikes and bus ripple.
- Verified in PSIM and a hardware prototype, improving pulse current compensation and lowering power loss.